YAN LUO （骆彦）

Ph.D. Student in Astrophysics | Peking University
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EDUCATION

Peking University, Beijing, China — Sep. 2024–Present
Ph.D. in Astrophysics, School of Physics — Expected 2029
Advisor: Gregory J. Herczeg

Sun Yat-sen University, Zhuhai, China — Sep. 2020–Jun. 2024
B.S. in Astronomy, School of Physics and Astronomy

RESEARCH INTERESTS

Pre-main-sequence stellar evolution; star formation; stellar activity and starspots; stellar ages

RESEARCH

Pre-Main-Sequence Evolution, Starspots, and Lithium — Oct. 2024–Present
Peking University | Advisor: Gregory J. Herczeg

- Investigate the effects of stellar surface inhomogeneities on lithium absorption, effective temperatures, luminosities, and inferred properties of young stars.
- Combined multi-epoch CFHT/ESPaDOnS spectroscopy with ASAS-SN photometry to measure rotational modulation of Li I 6708 Å and TiO absorption in young stars.
- Developed empirical two-temperature stellar-surface models to quantify the contribution of starspots to the lithium equivalent-width dispersion in NGC 2264.

Intrinsic Colors and Interstellar Extinction from APOGEE DR17 — Mar. 2022–Sep. 2023
Sun Yat-sen University | Advisors: He Zhao and Biwei Jiang

- Derived intrinsic color–effective-temperature relations for APOGEE DR17 giants and dwarfs using the blue-edge method and multi-band photometry.
- Cross-matched the stellar sample with Gaia DR3 and investigated systematic differences in intrinsic colors relevant to extinction measurements.

Radio Galaxy Zoo–EMU — Aug. 2021
Sun Yat-sen University | Advisor: Hongming Tang

- Processed ASKAP radio and WISE infrared imaging and developed Python workflows for preparing and deploying radio-source classification data to Zooniverse.
- Contributed data products and classifications used in subsequent studies of radio-galaxy morphology and anomalous sources in the EMU survey.

PUBLICATIONS

Refereed Articles

Luo, Y., Herczeg, G. J., Fang, M., Sun, Q., & Fu, X. 2026, "The Contribution of Starspots to the Lithium Spread in NGC 2264," *The Astrophysical Journal*, accepted. arXiv:2609.19653.

Bowles, M., Tang, H., Vardoulaki, E., Alexander, E. L., **Luo, Y.**, et al. 2023, "Radio Galaxy Zoo EMU: Towards a Semantic Radio Galaxy Morphology Taxonomy," *Monthly Notices of the Royal Astronomical Society*, **522**, 2584–2600. doi:10.1093/mnras/stad1021.

Segal, G., Parkinson, D., Norris, R. P., Hopkins, A. M., Andernach, H., Alexander, E. L., Carretti, E., Koribalski, B. S., Legodi, L. S., Leslie, S., **Luo, Y.**, et al. 2023, "Identifying Anomalous Radio Sources in the EMU Pilot Survey Using a Complexity-based Approach," *Monthly Notices of the Royal Astronomical Society*, **521**, 1429–1447. doi:10.1093/mnras/stad537.

Submitted Manuscripts

Tang, H., Yue, S., Wang, Z., Lai, J., Wei, L., **Luo, Y.**, et al., "Can I Trust You?: Interpreting Radio Galaxy Classifier with FR-DEEP Dataset and eXplainable AI," submitted to *The Astrophysical Journal Supplement Series*.

Conference Proceedings

Vardoulaki, E., Tang, H., Bowles, M., Segal, G., Mandhai, S., Alexander, E. L., Williams, W., **Luo, Y.**, et al. 2026, "Radio Galaxy Zoo EMU: Harnessing Citizen Science and AI to Advance Open Science Catalogues," *Proceedings of the International Astronomical Union*, Symposium 397. doi:10.1017/S1743921325104614.

HONORS & AWARDS

Peking University Third-Class Scholarship 2025

National Scholarship for Undergraduate Students 2021

Bronze Medal, 24th International Astronomy Olympiad 2019

OUTREACH & SCIENCE COMMUNICATION

Translator, "Finding the Oddities in the Night Sky" Feb. 2023
Original article by David Parkinson and Gary Segal; translated by **Yan Luo** and Hongming Tang.
Published in *Astronomy Enthusiast*, 2023, No. 2, 46–51.

TECHNICAL SKILLS

Programming Python, C